

AI LEGAL™

Law Firm Enterprise Workspace & Onboarding System

Complete Architecture, Onboarding Lifecycle, Multi-Tenant Switching & Operational Guide

Document Area	Details
Primary System	AI LEGAL™ Law Firm Enterprise Workspace
Core Functions	Enterprise onboarding, workspace creation, invitations, switching and tenant isolation
Primary Users	Advocates, Firm Partners, Associates, Consultants and Paralegals
Architecture	React frontend + Node/Express backend + MongoDB + JWT authorization

AI LEGAL — Law Firm Enterprise Workspace & Onboarding System

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1. Executive Summary & Overview

In high-volume legal practice, independent advocates, boutique law firms, and multi-partner corporate legal enterprises require distinct operational environments. The \\AI LEGAL Law Firm Enterprise Workspace & Onboarding System\\ enables law firms to transition from single-lawyer accounts into a collaborative, role-governed enterprise environment.

The system addresses the fundamental challenges of modern law firm management:

- \\Centralized Firm Identity:\\ Establishing a registered entity workspace with specific practice areas, bar council affiliations, and institutional branding.
- \\Multi-Tenant Workspace Switching:\\ Allowing senior partners and legal consultants to seamlessly switch between multiple firm workspaces, independent advocate profiles, and consulting desks without logging out.
- \\Secure Onboarding Pipeline:\\ Structured invitation handling for incoming partners, associates, and paralegals, complete with cryptographic tokens, real-time pending invitation resolution, and automatic role bootstrapping.
- \\Enterprise Data Segregation:\\ Strict tenant isolation ensuring that firm clients, privileged case files, legal strategies, and financial records remain strictly compartmentalized within the designated firm workspace.

2. Technical Architecture & Component Hierarchy

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User([Advocate / Firm Partner User]) --> Auth[JWT Authorization & Token Verification]

Auth --> ModeCheck{Has Active Firm Workspace?}

ModeCheck -- No --> OnboardingView[LawFirmOnboardingView.jsx]

ModeCheck -- Yes --> FirmDashboard[LawFirmDashboardSection.jsx]

OnboardingView --> CreateWs[Create Enterprise Firm Modal]

OnboardingView --> PendingInvites[Pending Firm Invitations Drawer]

CreateWs --> APIPostWs[POST /api/workspaces]

PendingInvites --> APIAcceptInvite[POST /api/workspaces/invitations/:id/accept]

PendingInvites --> APIRejectInvite[POST /api/workspaces/invitations/:id/reject]

APIPostWs --> MongoWS[(MongoDB Workspaces Collection)]

APIAcceptInvite --> MongoWS

APIPostWs --> LocalStorage[Set AILEGALLASTACTIVEWORKSPACEID]

APIAcceptInvite --> LocalStorage

LocalStorage --> EventTrigger[window.dispatchEvent 'workspacechanged']

EventTrigger --> FirmDashboard

\\

Key Frontend & Backend Source Files

- \\Frontend Onboarding View:\\ \\d:\\AI

Legal\\AILegalApp-backend-webpage-\\AI-LegalAppWebapp\\src\\Components\\LawFirmOnboardingView.jsx\\

- \\Frontend Dashboard Section:\\ \\d:\\AI

Legal\\AILegalApp-backend-webpage-\\AI-LegalAppWebapp\\src\\Components\\LawFirmDashboardSection.jsx\\

```
- \\Workspace Navigation & State Management:\\ \\d:\\AI
Legal\\AI\\LegalApp-backend-webpage-\\AI-LegalAppWebapp\\src\\pages\\HomeDashboard.jsx\\

- \\Backend Workspace Controller:\\ \\d:\\AI
Legal\\AI\\LegalApp-backend-webpage-\\AI-LegalAppBACKend\\controllers\\workspaceController.js\\

- \\Backend Workspace Router:\\ \\d:\\AI
Legal\\AI\\LegalApp-backend-webpage-\\AI-LegalAppBACKend\\routes\\workspaceRoutes.js\\

- \\Backend Workspace Model:\\ \\d:\\AI
Legal\\AI\\LegalApp-backend-webpage-\\AI-LegalAppBACKend\\models\\Workspace.js\\
```

3. Law Firm Enterprise Onboarding Flow

When a user switches their application mode to \\Law Firm\\ (\\role === 'lawfirm'), the application verifies if the user belongs to at least one active workspace or has pending invites. If no active workspace is resolved, the system mounts \\LawFirmOnboardingView.

3.1 Step 1: Practice Area Specialization Selection

Firms configure their core litigation domains from the standard Indian legal practice taxonomy:

- \\Civil Litigation: Contract disputes, property suits, recovery, injunctions.
- \\Criminal Defence: Bail applications, trial advocacy, economic offences, IPC/BNS compliance.
- \\Corporate & Commercial: M&A, shareholder agreements, NCLT insolvency proceedings.
- \\Property & Real Estate: Title verification, land acquisition, RERA disputes.
- \\Family Law: Divorce, child custody, maintenance under HMA & Special Marriage Act.
- \\Taxation & Banking: GST litigation, ITAT appeals, DRT recovery proceedings.
- \\Labour & Employment: Industrial disputes, POSH compliance, employment contracts.
- \\Arbitration & ADR: Domestic and international commercial arbitration under ACA 1996.
- \\IPR & Technology: Trademark registration, patent infringement, cyber law compliance.
- \\Consumer Disputes: District, State, and National Consumer Commission (NCDRC) litigation.

3.2 Step 2: Firm Creation Payload & API Transmission

The managing partner enters the formal law firm name (e.g., \\Singhania & Partners Chambers\\):

```
\\javascript
// From LawFirmOnboardingView.jsx (Lines 71-106)
const handleCreateFirm = async (e) => {
  e?.preventDefault();
  if (!firmName.trim()) {
    toast.error('Law Firm Name is required.');
```

```
    return;
  }
  try {
    setIsSubmitting(true);
    const tid = toast.loading('Creating Enterprise Law Firm Workspace...');
    const res = await apiService.request('/workspaces', {
      method: 'POST',
      data: {&#x20;
        name: firmName.trim(),
```

```

practiceAreas: selectedPracticeAreas
}
});
const data = res?.data || res;
if (data && data.success && data.workspace) {
  const newWs = data.workspace;
  const newWsId = newWs.\id || newWs.id;
  // Store in LocalStorage for persistence across tab reloads
  localStorage.setItem('AILEGALLASTACTIVEWORKSPACEID', newWsId);
  toast.success(\Welcome to ${firmName}! Firm Workspace Created.\, { id: tid });
  setIsCreateModalOpen(false);
  if (onWorkspaceCreated) onWorkspaceCreated(newWs);
}
} catch (err) {
  toast.error(err?.response?.data?.error || 'Failed to create workspace.');
```

```

} finally {
```

```

  setIsSubmitting(false);
```

```

}
```

```

};
```

```

\\
```

3.3 Step 3: Backend Workspace Genesis & Owner Initialization

On the backend (\workspaceController.js), creating a workspace executes the following operations:

1. Validates that the requesting advocate has an active session (\req.user.id).
2. Creates the \Workspace\ entity with \type: 'lawfirm\' and initializes the owner with the maximum administrative role (\Firm Partner (Owner)).
3. Automatically sets seat caps:
 - \maxSeats\: Defaults to 10 for Standard Firm Tier (expandable up to 100+ on Enterprise).
 - \usedSeats\: Initialized to 1 (the creating partner).
4. Creates the audit trail record in \WorkspaceActivity.
5. Returns the fully hydrated workspace object to the frontend.

4. Multi-Tenant Workspace Switching Engine

Advocates in India often collaborate across different entities:

- As \Managing Partner\ in their proprietary firm.
- As \Senior Consultant\ in an allied corporate advisory firm.
- As \Of-Counsel\ on independent high-court briefing dockets.

4.1 Persistence Mechanism: \AILEGALLASTACTIVEWORKSPACEID\

The client application synchronizes the active firm context across all components using \localStorage\ key \AILEGALLASTACTIVEWORKSPACEID.

```

\\
```

localStorage:

Key: "AILEGALLASTACTIVEWORKSPACEID"

Value: "66f4b23a9d18c4e098a5ef12" (MongoDB ObjectId of the active Firm)

\\

4.2 Cross-Component Event Synchronization

When a partner selects a different firm from the workspace dropdown switcher:

1. `\localStorage.setItem('AILEGALLASTACTIVEWORKSPACEID', newWorkspaceId)` is updated.
2. A custom browser window event `\workspacechanged` is broadcast:

\\javascript

```
window.dispatchEvent(new Event('workspacechanged'));
```

\\

3. All mounted workspaces, case pipelines, document docket, and calendar views immediately detect the event, invalidate their cached queries, and re-fetch records scoped strictly to the new `\workspaceId`:

\\javascript

// LawFirmDashboardSection.jsx (Lines 23-37)

```
useEffect(() => {
  const handleWorkspaceChange = () => {
    const current = localStorage.getItem('AILEGALLASTACTIVEWORKSPACEID');
    setActiveWsId(current);
    if (typeof onRefresh === 'function') {
      onRefresh();
    }
  };
  window.addEventListener('workspacechanged', handleWorkspaceChange);
  window.addEventListener('userrolechanged', handleWorkspaceChange);
  return () => {
    window.removeEventListener('workspacechanged', handleWorkspaceChange);
    window.removeEventListener('userrolechanged', handleWorkspaceChange);
  };
}, [onRefresh]);
```

\\

5. Joining an Existing Firm: Invitation Lifecycle

Lawyers who are invited to join an existing law firm do not need to construct a new workspace. Instead, `\LawFirmOnboardingView` proactively polls the pending invitations endpoint.

5.1 Invitation Retrieval (`\GET /api/workspaces/invitations/pending`)

Upon mounting, the component contacts the backend using the authenticated advocate's verified email and phone number:

\\javascript

```
const fetchPendingInvites = async () => {
  setIsFetchingInvites(true);
  const res = await apiService.request('/workspaces/invitations/pending');
```

```

const data = res?.data || res;
if (data && data.success && Array.isArray(data.invitations)) {
  setPendingInvites(data.invitations);
}
};
\\

```

5.2 Accept Invitation Protocol (\POST /api/workspaces/invitations/:id/accept)

When the invitee clicks \\"Join Firm"\\:

1. The backend verifies that the firm has not exceeded its \maxSeats\ limit.
2. Adds the user's \userId\ to the firm's \members\ collection with the assigned designation (e.g. \Associate Advocate) and permissions.
3. Sets \invitation.status = 'Accepted'.
4. The client sets the accepted workspace as the active workspace in \localStorage\ and triggers a full dashboard refresh.

\\javascript

```

const handleAcceptInvite = async (inviteId, workspaceId) => {
  const tid = toast.loading('Accepting invitation...');
  const res = await apiService.request(/workspaces/invitations/${inviteId}/accept, {
    method: 'POST'
  });
  if (res?.data?.success) {
    toast.success('Joined Law Firm Workspace successfully!', { id: tid });
    if (workspaceId) {
      localStorage.setItem('AILEGALLASTACTIVEWORKSPACEID', workspaceId);
    }
    if (onWorkspaceCreated) onWorkspaceCreated();
  }
};
\\

```

5.3 Reject / Decline Protocol (\POST /api/workspaces/invitations/:id/reject)

If an advocate declines an invitation:

- The backend marks the invitation token as \Declined'.
- The item is removed from the advocate's pending list without alerting other team members unnecessarily.

6. End-to-End API Reference

\ Endpoint Method Auth Required Description
\:--- :--- :--- :---
\ /api/workspaces\ \GET\ Bearer JWT Retrieves all firm workspaces where the current user is an Owner, Partner, or Member.
\ /api/workspaces\ \POST\ Bearer JWT Creates a new enterprise firm workspace. Body: \{ name, practiceAreas \}.

`\| /api/workspaces/invitations/pending\ | \GET\ | Bearer JWT | Returns all unaccepted invitations addressed to the logged-in user's email/phone. |`

`\| /api/workspaces/invitations/:id/accept\ | \POST\ | Bearer JWT | Accepts an invitation and binds the advocate into the firm roster. |`

`\| /api/workspaces/invitations/:id/reject\ | \POST\ | Bearer JWT | Declines an invitation and revokes the joining token. |`

`\| /api/workspaces/:workspaceId/members\ | \GET\ | Bearer JWT | Returns the full roster of active and pending members in the designated firm. |`

7. Data Models & Schemas

Workspace Schema (\models/Workspace.js)

`\\typescript`

`interface IWorkspace {`

`\id: string;`

`name: string;`

`type: 'lawfirm' | 'advocate' | 'enterprise';`

`ownerId: string;`

`practiceAreas: string[];`

`maxSeats: number; // Default 10`

`usedSeats: number; // Dynamically computed`

`settings: {`

`allowAssociateDirectClientChat: boolean;`

`requirePartnerDraftApproval: boolean;`

`enableMultiAdvocateHearingsCalendar: boolean;`

`};`

`createdAt: Date;`

`updatedAt: Date;`

`}`

`\\`

8. Troubleshooting & Frequently Asked Questions (FAQ)

Q1: An advocate accepted an invitation but still sees the Onboarding screen. Why?

`\\Cause:\\ The browser's \AILEGALLASTACTIVEWORKSPACEID\ was not updated to point to the newly joined workspace, or cached user claims have not refreshed.`

`\\Resolution:\\ The advocate should click the \"Switch Workspace\" dropdown in the navigation header or hard-refresh the page (\Ctrl + F5), which forces \HomeDashboard.jsx\ to re-fetch /api/workspaces\ and auto-select the accepted firm ID.`

Q2: Can a firm partner create more than one Law Firm Workspace?

`\\Yes.\\ Senior advocates running independent corporate and criminal practices can create distinct workspaces (e.g. \"Apex Tax Chambers\" and \"Apex Criminal Defence\"). The workspace switcher allows toggling between them instantly.`

Q3: What happens if an advocate changes their registered email?

Invitations are linked to the specific email entered during the invite. If an associate's email changes, the Managing Partner must issue a new invite to their updated email address via `\InviteTeamMemberModal`.